

Worksheet 12

Single-phase induction motor

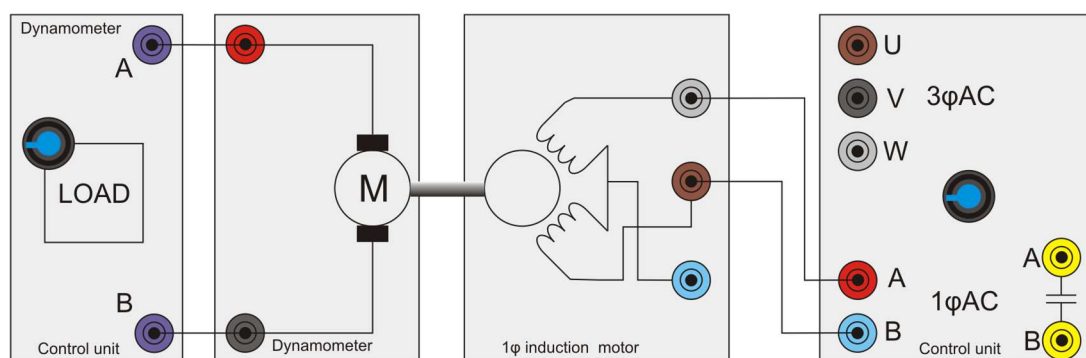
Modern electrical machines



One problem with series and shunt motors is that they use 'brushes' in the rotor circuit. These wear out and need replacing from time to time.

Induction motors have no brushes and so require less maintenance.

The photograph shows an induction motor and its fitted gearbox.



Over to you:

- 1) Set up the single phase induction motor as shown in the diagram above.
- 2) On your PC run 'Open Control_1Phase.bat'
- 3) Connect the windings to terminals **A** and **B**.
When the software application is running, these give a single phase output signal.
- 4) However, you will find that the motor does not start.

Oh dear!

So what?

You have just discovered a problem with the induction motor - it needs something to get it started.

When the induction motor is powered, a little physical nudge in one direction often starts it. Unfortunately, in this rig, you cannot touch the shaft, for safety reasons.

Another problem is that it does not matter which direction you nudge the motor in - it can rotate either way.

Next we will look at how to make the system more predictable.